Fiscal Year:	FY 2026	**Task Last Updated:**	FY 08/18/2025
PI Name:	Greenhalgh, Preston C. M.S.		
Project Title:	HF Occupant Protection		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **Dynamic Loads**:Risk of In-Mission Injury and Performance Decrements and Long-term Health Effects due to Dynamic Loads		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	preston.c.greenhalgh@nasa.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	281-483-6639
Organization Name:	KBR Wyle/NASA Johnson Space Center		
PI Address 1:	JSC B21		
PI Address 2:	2400 NASA Parkway		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2015	**End Date:**	10/01/2031
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	2025 Update: The Institutional Review Board (IRB) protocol was updated in August 2025 to reflect the addition of Katherine Hoerner and Parker Berthelson as Co-Is and the removal of Jacob Putnam and Jeff Somers as Co-Is. Teresa Reiber was removed as a Co-I because she took a new position between the time of the proposal submission and the funding award. (Ed., 8/16/25).		
COI Name (Institution):	Yates, Keegan Ph.D. (KBR/NASA Johnson Space Center) Newby, Nathaniel M.S. (KBR/NASA Johnson Space Center) Drake, Aaron M. M.S. (KBR/NASA Johnson Space Center) Howes, Caroline E. M.S. (KBR/NASA Johnson Space Center) Thompson, Rachel L. Ph.D. (KBR/NASA Johnson Space Center) Berthelson, Parker (Aegis Aerospace Inc.) Hoerner, Katherine (Aegis Aerospace Inc.)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	October 2024 - Updated Task Description per PI (Ed., 10/11/24) Currently the impact load imparted to crewmembers landing in the Soyuz vehicle is unknown. This study is the first systematic assessment of the number and types of injuries associated with Soyuz landing. To date, we have found that nearly a third of United States Orbital Segment (USOS) crewmembers are experiencing injuries. Most of these injuries are minor, but they exceed expected rates based on analysis of seat accelerometer data from airborne and drop tests of the vehicle. The yet to be answered question is whether spaceflight deconditioning renders crewmembers more susceptible to landing impact injuries. Another possibility is that the Soyuz landing load is higher than our current estimates. It could also be that our analytical tools are insufficient to predict injury rates accurately for space vehicles. A final possibility is that some combination of these factors are responsible. Current occupant protection requirements levied upon future NASA space vehicles are based, in part, on the mathematical prediction that minor injuries due to Soyuz landings have less than a 5% probability of occurrence. Understanding the risk of injury during Soyuz landings and the validity of current crew risk models will allow for improved crew protection in future spacecraft designs. The following are the specific aims for this task: Aim 1: This aim consists of data collection from two sources. One is flight medical records from a database maintained by the NASA Lifetime Surveillance of Astronaut Health (LSAH). This data is only obtainable for US astronauts. The other data source is from a survey that crewmembers are asked to complete. The survey requires an additional consent process. Aim 2: The study team has not had success obtaining Soyuz seat acceleration data, and NASA management has abandoned any future effort of obtaining it. However, NASA is supporting a Small Business Innovation Research (SBIR) grant for the development of miniaturized accelerometers that can be worn by astronauts during dynamic phases of spaceflights. The conceptual design would allow the crew to adhere these sensors to locations such as the head or spine. Aim 3: An analysis has been performed to identify the expected rates of Soyuz injury using the Brinkley model. These expected rates are compared with the actual injury rates uncovered by this study.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research investigates the injury risk associated with spaceflight. While there is always risk for spaceflight, this research attempts to better understand why NASA astronauts and other crewmembers sustain more injuries than expected from Soyuz landings and how this relates to other current and future spaceflight vehicles.
Task Progress:	This study began during time period in which Soyuz was the only vehicle transporting NASA and United States Operational Segment (USOS) crewmembers back and forth from the International Space Station (ISS). Within the last several years, NASA and USOS crewmembers have been flying on the SpaceX Crew Dragon vehicle as part of the Commercial Crew Program (CCP). This study has been expanded to include NASA and USOS crewmembers flying on Commercial Crew Program (CCP) missions as well as private astronaut missions (PAM). Previous reports outline the major activities for completing Aims 2 and 3. This reporting period continues to address Aim 1. This aim consists of data collection from two sources. One is flight medical records from a database maintained by the NASA Lifetime Surveillance of Astronaut Health (LSAH). This data is only obtainable for NASA astronauts. The other data source is from a survey that crewmembers are asked to complete. The survey can be completed by NASA and USOS astronauts, spaceflight participants, and PAM crewmembers. The survey requires an additional consent process and collects any injuries that were attributable to landing dynamics. Injuries identified upon landing are classified using the Operationally Relevant Injury Scale (ORIS) with ratings from Class 0–Class IV. ORIS is an adaptation of an injury classification scale developed by the American Association of Automotive Medicine called the Abbreviated Injury Scale (AIS). The ORIS was developed to address NASA's need for a scale that not only categorizes injury by severity, but also accounts for an astronaut's ability to operate and carry out the mission in a spaceflight environment and contains a return to flight status component. Only injuries that are associated with the re-entry, descent, and landing are counted in the data below. Pre-existing injuries (e.g., injuries from on-station exercise prior to undock) or non-dynamic related injuries (e.g., suit ingress) are not counted. This is the first year that has incorporated data collected from SpaceX's Crew Dragon into the Soyuz data. Through MS-26 for Soyuz and Crew-10 for Dragon, a total of 130 data collections have been performed: 84 Soyuz and 46 Dragon.
Bibliography Type:	Description: (Last Updated: 05/20/2026)
Conference Materials (Downloadable)	Greenhalgh PC, Yates KM, Reiber TM, Newby NJ. "Soyuz landing risk characterization." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 SOYUZ LANDING RISK CHARACTERIZATION [SESS-655].pdf